MSG No. 129.422 P_I4/nmm [Type IV, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4/m'mm

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[1,2,3,6,12,13,15,16,20,21,23,24,25,26,27,30]
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,5,7,8,9,10,11,14,17,18,19,22,28,29,31,32]

Table 2: Wyckoff site: 2b, site symmetry: 4/m'mm

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,2,3,6,12,13,15,16,20,21,23,24,25,26,27,30]
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[4,5,7,8,9,10,11,14,17,18,19,22,28,29,31,32]

Table 3: Wyckoff site: 4c, site symmetry: m'mm.

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[1,6,12,13,20,21,25,30]
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[2,3,15,16,23,24,26,27]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[4,5,9,14,17,22,28,29]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[7,8,10,11,18,19,31,32]

Table 4: Wyckoff site: 4d, site symmetry: -4m2

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[1,6,7,8,10,11,12,13]
2	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[2,3,4,5,9,14,15,16]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[17,22,23,24,26,27,28,29]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[18,19,20,21,25,30,31,32]

Table 5: Wyckoff site: 4e, site symmetry: 4mm

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[1,2,3,6,12,13,15,16]
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[4,5,7,8,9,10,11,14]
3	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[17,18,19,22,28,29,31,32]
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	[20,21,23,24,25,26,27,30]

Table 6: Wyckoff site: 8f, site symmetry: ..2/m

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[1,8,9,16]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,4,10,12]
3	$\left[\frac{1}{2}, 0, 0\right]$	[3,5,11,13]

continued ...

Table 6

No.	position	mapping
4	$[0,\ 0,\ 0]$	`[6,7,14,15]`
5	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[17,24,25,32]`
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[18,20,26,28]`
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[19,21,27,29]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[22,23,30,31]`

Table 7: Wyckoff site: `8g`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[1,6,12,13]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[2,3,15,16]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[4,5,9,14]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[7,8,10,11]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[17,22,28,29]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[18,19,31,32]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[20,21,25,30]`
8	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[23,24,26,27]`

Table 8: Wyckoff site: `8h`, site symmetry: `m'.2'm`

No.	position	mapping
1	$\left[x,\ x,\ \frac{3}{4}\right]$	`[1,16,23,30]`
2	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	`[2,12,21,27]`
3	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[3,13,20,26]`
4	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	`[4,10,19,29]`
5	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[5,11,18,28]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[6,15,24,25]`
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[7,14,17,32]`
8	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[8,9,22,31]`

Table 9: Wyckoff site: `8i`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,13,20,30]`
2	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[2,16,23,27]`
3	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[3,15,24,26]`
4	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[4,14,17,29]`
5	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[5,9,22,28]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[6,12,21,25]`
7	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[7,11,18,32]`
8	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	`[8,10,19,31]`

Table 10: Wyckoff site: `8j`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,13,20,30]`
2	$\left[\frac{3}{4},\ x,\ \frac{3}{4}\right]$	`[2,16,23,27]`
3	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[3,15,24,26]`
4	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[4,14,17,29]`
5	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,9,22,28]`
6	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[6,12,21,25]`
7	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[7,11,18,32]`
8	$\left[\frac{1}{4},\ -x,\ \frac{1}{4}\right]$	`[8,10,19,31]`

Table 11: Wyckoff site: `16k`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,7]`
2	$\left[-x,\ x,\ 0\right]$	`[2,5]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[3,4]`
4	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	`[6,8]`
5	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[9,15]`
6	$\left[x,\ -x,\ 0\right]$	`[10,13]`
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[11,12]`
8	$\left[x+\frac{1}{2},\ x,\ 0\right]$	`[14,16]`
9	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[17,23]`
10	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[18,21]`
11	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[19,20]`
12	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[22,24]`
13	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	`[25,31]`
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[26,29]`
15	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[27,28]`
16	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[30,32]`

Table 12: Wyckoff site: `16l`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{3}{4}\right]$	`[1,30]`
2	$\left[\frac{1}{2}-y,\ x,\ \frac{3}{4}\right]$	`[2,27]`
3	$\left[y,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[3,26]`
4	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{4}\right]$	`[4,29]`
5	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[5,28]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[6,25]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[7,32]`
8	$\left[-y,\ -x,\ \frac{1}{4}\right]$	`[8,31]`
9	$\left[-x,\ -y,\ \frac{1}{4}\right]$	`[9,22]`
10	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	`[10,19]`
11	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[11,18]`
12	$\left[\frac{1}{2}-x,\ y,\ \frac{3}{4}\right]$	`[12,21]`

continued ...

Table 12

No.	position	mapping
13	$\left[x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[13,20]
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[14,17]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[15,24]
16	$\left[y,\ x,\ \frac{3}{4}\right]$	[16,23]

Table 13: Wyckoff site: 16m, site symmetry: ..m

No.	position	mapping
1	$[x,\ x,\ z]$	[1,16]
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[2,12]
3	$\left[x,\ \frac{1}{2}-x,\ z\right]$	[3,13]
4	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	[4,10]
5	$\left[-x,\ x+\frac{1}{2},\ -z\right]$	[5,11]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	[6,15]
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[7,14]
8	$[-x,\ -x,\ -z]$	[8,9]
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17,32]
10	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[18,28]
11	$\left[x+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[19,29]
12	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[20,26]
13	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-z\right]$	[21,27]
14	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	[22,31]
15	$\left[x,\ x,\ \frac{1}{2}-z\right]$	[23,30]
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[24,25]

Table 14: Wyckoff site: 16n, site symmetry: .m.

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	[1,12]
2	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ z\right]$	[2,15]
3	$\left[y,\ \frac{1}{4},\ z\right]$	[3,16]
4	$\left[\frac{3}{4},\ -y,\ -z\right]$	[4,9]
5	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	[5,14]
6	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	[6,13]
7	$\left[y+\frac{1}{2},\ \frac{3}{4},\ -z\right]$	[7,10]
8	$\left[-y,\ \frac{3}{4},\ -z\right]$	[8,11]
9	$\left[\frac{3}{4},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17,28]
10	$\left[-y,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[18,31]
11	$\left[y+\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[19,32]
12	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[20,25]
13	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-z\right]$	[21,30]
14	$\left[\frac{3}{4},\ -y,\ z+\frac{1}{2}\right]$	[22,29]
15	$\left[y,\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[23,26]
16	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[24,27]

Table 15: Wyckoff site: 32o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{2}-y,\ x,\ z\right]$	[2]
3	$\left[y,\ \frac{1}{2}-x,\ z\right]$	[3]
4	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	[4]
5	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[5]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[6]
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[7]
8	$[-y,\ -x,\ -z]$	[8]
9	$[-x,\ -y,\ -z]$	[9]
10	$\left[y+\frac{1}{2},\ -x,\ -z\right]$	[10]
11	$\left[-y,\ x+\frac{1}{2},\ -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ y,\ z\right]$	[12]
13	$\left[x,\ \frac{1}{2}-y,\ z\right]$	[13]
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[14]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[15]
16	$[y,\ x,\ z]$	[16]
17	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17]
18	$\left[-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[19]
20	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[20]
21	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[21]
22	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[22]
23	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[23]
24	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[24]
25	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[25]
26	$\left[y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-y,\ x,\ \frac{1}{2}-z\right]$	[27]
28	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[28]
29	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[29]
30	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[30]
31	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[31]
32	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[32]